

Post-Hospital Recovery Management and Early-Warning System

Project Proposal for UCS503P
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1 Higher-order goal

... secondary framing

This project aims to improve the continuity and organization of patient care after hospital discharge by providing a centralized software platform for monitoring recovery progress, managing post-discharge activities, and identifying unusual recovery patterns that may require professional attention.

The immediate engineering objective is to translate post-hospital recovery management into a measurable, deployable software solution that connects patients, healthcare professionals, and recovery data through a structured workflow.

The system will not attempt to diagnose diseases or replace healthcare professionals. Instead, it will provide decision-support features that organize patient-reported information, monitor recovery trends, and highlight potentially concerning deviations for professional review.

The project combines conventional software engineering with a lightweight machine-learning component. The software platform will manage users, recovery plans, medications, exercises, appointments, daily logs, notifications, and historical records, while machine learning will be used to identify deviations from expected recovery patterns and classify recovery trajectories.

2 Time-to-value

... primary heuristic

We will deliver the project through incremental iterations, beginning with the core patient recovery workflow before introducing machine-learning functionality.

The first iteration will focus on establishing a usable system where:

- A healthcare professional can create a patient's post-discharge recovery plan.
- A patient can access the plan and submit daily recovery information.
- Medication, exercise, and appointment schedules can be tracked.
- Recovery information is stored with timestamps.
- Patients can view their recovery history and progress.
- Healthcare professionals can view patient recovery information.

Subsequent iterations will introduce:

- Recovery scoring.
- Recovery trend visualization.
- Rule-based alerts.
- Machine-learning-based trajectory analysis.
- Risk/deviation indicators.
- Healthcare-professional dashboards.
- Notifications.

We will prioritize working software and measurable functionality over premature complexity. Weekly iterations will be used to obtain feedback and identify problems in the workflow early.

Success will be measured not only by whether the ML model produces predictions, but also by whether the complete software system can reliably collect recovery information, identify changes, present them clearly, and support timely professional review.

3 Problem Statement

After a patient leaves a hospital, recovery continues for days or weeks, but the patient’s recovery information is often fragmented between prescriptions, appointments, personal notes, and occasional follow-up consultations.

Common problems include:

- **Limited continuous monitoring:** Healthcare professionals may have little visibility into the patient’s recovery between scheduled appointments.
- **Fragmented information:** Medication schedules, exercise plans, symptoms, pain levels, mobility, and appointments may be tracked separately.
- **Poor visibility of recovery trends:** A single measurement may not indicate whether a patient is improving or deteriorating. Changes over several days can be more informative.
- **Delayed identification of unusual patterns:** A patient may experience a gradual deviation from their expected recovery pattern without recognizing its significance.
- **Medication and exercise adherence:** Patients may forget or inconsistently follow prescribed post-discharge activities.
- **Follow-up coordination:** Appointments and recovery milestones can be difficult to manage alongside daily recovery tasks.
- **Information overload for healthcare professionals:** If large amounts of patient data are collected, professionals need a way to prioritize patients requiring attention rather than manually inspecting every record.
- **Lack of structured historical data:** Recovery information collected from patients is often not organized into a consistent longitudinal record.

Why this matters now: post-hospital care involves a period where the patient is responsible for a significant part of their recovery while professional interaction may be intermittent. A software system that organizes this period into a continuous and measurable workflow can reduce information gaps and help professionals identify patients whose recovery requires review.

4 Proposed Solution

... *Software Proposition*

4.1 Overview

Build a web-based Post-Hospital Recovery Management and Early-Warning System with the following components:

- **Patient portal:** Patients view their recovery plan, medications, exercises, appointments, and daily tasks.
- **Daily recovery logging:** Patients record structured information such as pain, mobility/function, sleep, symptoms, medication adherence, and exercise completion.
- **Recovery timeline:** The system displays the patient's recovery progress over time instead of only showing the latest measurement.
- **Recovery scoring:** A transparent application-level score summarizes multiple recovery dimensions.
- **Rule-based monitoring:** Explicitly configured thresholds and repeated changes generate alerts for review.
- **ML-based trajectory analysis:** A machine-learning model analyzes combinations and trends in recovery data to identify patterns that deviate from expected recovery trajectories.
- **Healthcare-professional dashboard:** Professionals can view patient status, trends, alerts, and historical recovery information.
- **Medication and exercise management:** Recovery activities can be scheduled and adherence can be recorded.
- **Appointment management:** Follow-up appointments and recovery milestones can be stored and tracked.
- **Notification system:** Patients can receive reminders while healthcare professionals can receive high-priority review notifications.
- **Administrative dashboard:** Provides system-level information such as active patients, outstanding alerts, adherence statistics, and recovery trends.

The system is designed as a decision-support platform rather than an autonomous diagnostic system. ML predictions and alerts will indicate that a patient's data requires review rather than claiming that a patient has a particular disease or complication.

4.2 Core workflow

4.2.1 1. Recovery plan creation

A healthcare professional creates a recovery plan for a discharged patient.

The plan contains:

- Recovery category/procedure
- Expected recovery duration
- Recovery milestones
- Measurements to track

- Medication schedule
- Exercise/activity schedule
- Restrictions
- Follow-up appointments

4.2.2 2. Patient onboarding

The patient receives access to the recovery plan and can view their daily tasks.

4.2.3 3. Daily check-in

The patient submits structured information such as:

- Pain level
- Mobility/function level
- Sleep duration
- Temperature, when relevant to the selected recovery plan
- Medication adherence
- Exercise adherence
- Newly reported symptoms
- General recovery notes

The system records every submission with a timestamp.

4.2.4 4. Recovery analysis

The system calculates the patient's current recovery indicators and compares recent measurements with historical measurements and the expected recovery trajectory.

4.2.5 5. Rule-based monitoring

Explicitly configured conditions are checked.

For example:

- A measurement exceeds a configured threshold.
- A concerning symptom is reported.
- Medication is repeatedly missed.
- Recovery measurements change significantly over a short period.

These events can create a review flag.

4.2.6 6. Machine-learning analysis

The ML model analyzes the patient's longitudinal data and estimates whether the current trajectory is:

- On Track
- Delayed
- Potentially Concerning

The model may also produce a deviation/risk probability.

4.2.7 7. Professional review

Healthcare professionals see prioritized patients on their dashboard.

They can inspect:

- Current values
- Historical trends
- Recovery score
- ML output
- Reasons/features associated with the flag
- Medication adherence
- Exercise adherence
- Patient notes
- Previous alerts

The healthcare professional remains responsible for deciding whether further action is necessary.

4.2.8 8. Feedback

The outcome of professional review can be recorded in the system.

This information can later become labeled data for evaluating or improving future versions of the ML model.

5 Solution Approach

... Engineering focus

This is an engineering course project, so the primary focus will be on building a reliable software system around the recovery workflow rather than developing a novel medical ML algorithm.

The ML component will use established, interpretable machine-learning techniques on structured longitudinal data.

5.1 Recovery data model

...structured patient history

The system will represent recovery as a sequence of time-stamped observations rather than isolated values.

A daily recovery record may contain:

Feature	Example
Recovery day	Day 7
Pain	5/10
Mobility/function	62%
Sleep	6.5 hours
Medication adherence	100%
Exercise adherence	67%
Temperature	37.2°C
Reported symptoms	None
Patient note	Optional
Timestamp	Date/time

The exact measurements will depend on the selected recovery scenario.

The initial implementation will focus on a limited set of measurable variables rather than attempting to monitor every possible medical parameter.

5.2 Recovery scoring

...transparent logic

A preliminary recovery score will be implemented using a transparent weighted formula.

For example:

Recovery Score =

- 30% Mobility/Function
- 25% Pain-related indicator
- 15% Exercise adherence
- 15% Medication adherence
- 15% Sleep indicator

The weights are application-level prototype assumptions and will not be presented as medically validated clinical standards.

The purpose of this score is to provide an understandable summary of multiple recovery dimensions.

It will be complemented by the patient's individual trend rather than being used as a standalone medical decision.

5.3 Rule-based alerts

...deterministic safety layer

Rules will be used for conditions that can be explicitly specified.

For example:

- **IF** a configured critical symptom is reported → create high-priority review flag
- **IF** a monitored measurement exceeds its configured threshold → create monitoring alert
- **IF** medication adherence falls below a configured level → create adherence alert

- **IF** recovery measurements show repeated deterioration → increase monitoring priority

The rules will be configurable according to the selected recovery scenario.

The system will not autonomously diagnose the patient or prescribe treatment.

5.4 Machine-learning component

...non-research approach

Machine learning will focus on recovery trajectory classification and anomaly detection.

The initial model will use structured features such as:

- Current pain level
- Change in pain over previous days
- Mobility/function
- Change in mobility
- Sleep
- Medication adherence
- Exercise adherence
- Reported symptoms
- Recovery day
- Procedure/recovery category

The initial classification task will be:

- ON TRACK
- DELAYED
- POTENTIALLY CONCERNING

A lightweight model such as Random Forest will be evaluated first because it works well with tabular data and is relatively interpretable.

For anomaly detection, an approach such as Isolation Forest may be evaluated to identify recovery observations that differ substantially from learned patterns.

The project will compare the performance of the selected model against a simple rule-based baseline.

This allows the project to answer an important engineering question: Does adding machine learning provide useful information beyond a simple deterministic recovery-score/rule system?

5.5 Dataset strategy

...practical educational approach

The project will not depend on finding one perfect public post-discharge dataset.

The dataset strategy will use two sources where possible:

5.5.1 Public/real-world data

Relevant publicly available healthcare and patient-reported-outcome datasets will be investigated and used where their variables and licensing/access conditions are appropriate.

5.5.2 Synthetic longitudinal data

Because a clean dataset containing all required post-discharge variables may not be publicly available, a synthetic patient dataset will be generated for development and testing.

The synthetic data will represent different recovery trajectories, such as:

- Normal recovery
- Gradually improving recovery
- Delayed recovery
- Recovery with adherence problems
- Recovery with temporary deviations
- Recovery with significant deterioration patterns

The synthetic data generator will create time-series observations rather than independent random records.

This allows the system to contain realistic longitudinal relationships such as:

Expected recovery: Pain: $8 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 3$

Mobility: $30 \rightarrow 38 \rightarrow 45 \rightarrow 55 \rightarrow 65 \rightarrow 72$

while also generating patients whose trajectories deviate from the expected pattern.

Synthetic data will be clearly identified as simulated data and will not be presented as real clinical evidence.

5.6 Web architecture

... web-based solution

The system will use a 3-tier architecture:

5.6.1 Frontend

A responsive React application will provide:

- Patient dashboard
- Daily check-in
- Recovery timeline
- Medication tracker
- Exercise tracker
- Appointment view
- Healthcare-professional dashboard
- Alert management

5.6.2 Backend

A FastAPI backend will handle:

- Authentication
- Authorization
- Patient management
- Recovery plans
- Daily logs
- Medication schedules
- Exercise schedules
- Appointment management
- Alert generation
- ML prediction requests
- Notification logic
- Audit/history records

5.6.3 Database

The database will store:

- Users
- Patients
- Healthcare professionals
- Recovery plans
- Daily recovery logs
- Medications
- Exercises
- Appointments
- Alerts
- ML predictions
- Review outcomes

The architecture will keep the ML component separate from the primary API logic so that the model can be updated without redesigning the entire application.

5.7 ML service architecture

The ML pipeline will be separated into:

Patient Recovery Data → Data Validation → Feature Engineering → ML Model → Prediction → Recovery Engine → Database → Dashboard / Alert

The model will not directly communicate a medical diagnosis to the patient.

Instead, it will return structured information such as:

- status: “potentially_concerning”
- probability: 0.76
- model_version: “v1”
- timestamp: ...

The backend can then determine how that prediction should be displayed and whether professional review should be prioritized.

5.8 Authentication and authorization

Different user roles will have different permissions.

5.8.1 Patient

Can:

- view their recovery plan
- submit daily logs
- view their own recovery history
- manage reminders
- view appointments
- receive notifications

5.8.2 Healthcare professional

Can:

- create recovery plans
- view assigned patients
- inspect recovery trends
- review alerts
- record review outcomes
- modify recovery schedules

5.8.3 Administrator

Can:

- manage users
- monitor system activity
- view aggregate statistics
- manage system configuration

Role-based access control will prevent patients from accessing other patients' information.

6 Evaluation Criterion

...measurable and attributable

The project will define measurable criteria for both the software system and the ML component.

6.1 Primary evaluation metric

Recovery monitoring completeness

Measure the percentage of scheduled patient check-ins that are successfully recorded and processed by the system.

Target: $\geq 90\%$ of simulated scheduled check-ins successfully stored and processed during the pilot.

Attribution: Calculated from database records containing expected check-ins and successfully submitted logs.

6.2 Secondary software metrics

Alert processing time

Measure the time between submission of a recovery log and creation of the corresponding alert/prediction.

Target: The system should process a normal recovery submission within a few seconds under the project test load.

Medication adherence tracking

Measure whether the system correctly records scheduled and completed medication events.

Appointment tracking

Measure whether scheduled appointments and reminders are correctly generated and displayed.

System reliability

Measure:

- API availability
- failed requests
- database errors
- authentication failures

User clarity

Pilot users can rate how clearly the system communicates:

- recovery status
- daily tasks

- alerts
 - recovery trends
- using a simple 1–5 scale.

6.3 ML evaluation metrics

The ML model will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Special attention will be given to recall for the potentially concerning class, because failing to identify a concerning simulated recovery trajectory is an important failure mode for the prototype.

The model will be compared with a simple rule-based baseline.

Example evaluation:

	Rule System	ML Model
Accuracy	XX%	XX%
Precision	XX%	XX%
Recall	XX%	XX%
F1 Score	XX%	XX%

The final values will be filled after model training and testing rather than being claimed beforehand.

6.4 Pilot validation plan

A controlled educational pilot can be conducted using:

- 10–20 test users
- Simulated patient recovery cases
- Student testers acting as patients
- Student/teacher-supervised users acting as healthcare professionals where appropriate

The pilot will measure:

- Completion of daily check-ins
- Correctness of workflow
- Alert generation
- Dashboard usability
- ML prediction performance
- Response time
- User clarity

No real patient’s medical information will be required for the educational prototype.

7 Scalability

... with foresight

The proposition should scale in both theoretical and practical senses.

7.1 1. Scalable architecture

The system will support growth in patient and recovery-log volume through:

- Separate frontend and backend components
- Stateless API design
- Database indexing
- Pagination for patient records
- Efficient querying of time-series recovery logs
- Separate ML inference service
- Background processing for notifications

7.2 2. ML scalability

The ML model will be versioned independently from the main application.

For example:

Model v1 $\rightarrow$ Model v2 $\rightarrow$ Model v3

Predictions will store the model version that generated them.

This makes it possible to compare model versions and roll back an updated model if required.

7.3 3. Operational deployment

The system should be deployable using common web infrastructure:

- Managed database
- Cloud-hosted backend
- Cloud-hosted frontend
- Containerized services where appropriate
- CI/CD pipeline
- Environment-based configuration

8 Engine availability heuristic

A mature set of development tools and services is available to implement the system as a web-based project.

The proposed stack is:

- **Frontend:** React
- **Styling/UI:** Tailwind CSS
- **Backend:** FastAPI

- **Database:** PostgreSQL or MongoDB
- **ML:** Python, NumPy, pandas, scikit-learn
- **Data visualization:** Recharts
- **Authentication:** JWT/token-based authentication
- **Testing:** Pytest and frontend testing tools
- **Version control:** Git and GitHub
- **CI:** GitHub Actions
- **Deployment:** Vercel for frontend and Render or equivalent platform for backend
- **Containerization:** Docker where required

These are mature technologies, allowing the project team to focus on recovery workflow design, ML integration, testing, security, and system correctness rather than developing infrastructure from scratch.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

... first iteration

The first iteration will contain:

- Patient and healthcare-professional authentication
- Patient profile
- Recovery plan creation
- Patient recovery-plan view
- Daily recovery check-in
- Recovery data storage
- Medication schedule
- Exercise schedule
- Appointment schedule
- Basic recovery history
- Database timestamps
- Basic backend tests
- CI pipeline
- Deployment to staging

At the end of this iteration, a complete patient → daily check-in → database → professional dashboard workflow should be functional.

9.2 Subsequent deliverables

9.2.1 Iteration 2

- Recovery score
- Recovery trend graphs
- Medication adherence calculation
- Exercise adherence calculation
- Rule-based alerts
- Notification system

9.2.2 Iteration 3

- Synthetic patient-data generator
- ML training pipeline
- Recovery trajectory classification
- Anomaly detection
- Model evaluation
- Prediction API

9.2.3 Iteration 4

- Healthcare-professional command center
- Patient prioritization
- ML prediction visualization
- Alert history
- Professional review workflow
- Feedback/outcome recording

9.2.4 Final iteration

- Security improvements
- Performance testing
- Integration testing
- ML model comparison
- UI refinement
- Documentation
- CI/CD improvements
- Final deployment
- Final evaluation

10 Risks and mitigations

10.1 Medical interpretation risk

Risk: Users may interpret the ML output as a medical diagnosis.

Mitigation: The system will clearly state that ML outputs are decision-support indicators for professional review. The system will not provide autonomous diagnosis or treatment recommendations.

10.2 Data availability

Risk: A public dataset may not contain all required post-discharge variables.

Mitigation: Use relevant public datasets where appropriate and create a documented synthetic longitudinal dataset for development and evaluation.

10.3 Synthetic-data bias

Risk: A model trained primarily on synthetic data may not represent real-world patients.

Mitigation: Clearly identify synthetic data as simulated, evaluate the model only within the scope of the simulated experiment, and avoid making claims of clinical validity.

10.4 False alerts

Risk: The system may produce too many alerts.

Mitigation: Use a combination of deterministic rules and ML confidence thresholds, record alert history, and evaluate false-positive rates during testing.

10.5 Missed concerning patterns

Risk: The ML system may fail to identify a simulated concerning recovery pattern.

Mitigation: Evaluate recall separately for the potentially concerning class and keep the professional review layer in the workflow.

10.6 Data privacy

Risk: Recovery information is sensitive.

Mitigation: For the educational prototype, use synthetic data whenever possible. If real data is ever introduced, collect only necessary information, apply role-based access control, avoid exposing patient information through logs, and follow appropriate privacy requirements.

10.7 Model drift

Risk: A future dataset may differ from the training data.

Mitigation: Store model versions, prediction timestamps, and evaluation metrics. Future models will be evaluated before deployment.

10.8 System reliability

Risk: Failure of the ML service should not make the entire application unavailable.

Mitigation: The core patient-recording and recovery-management system will continue to function even if ML inference is temporarily unavailable. ML predictions will be treated as an additional service rather than the foundation of the entire application.

11 Summary

This project proposes a deployable web-based Post-Hospital Recovery Management and Early-Warning System that provides a structured workflow for managing the period after hospital discharge.

The system will allow healthcare professionals to create recovery plans and patients to record daily recovery information, manage medications and exercises, track appointments, and monitor their progress over time.

A transparent recovery-score and rule-based monitoring layer will provide understandable baseline functionality, while machine learning will analyze longitudinal recovery data to identify patterns that deviate from expected recovery trajectories.

The project deliberately avoids attempting autonomous medical diagnosis. Instead, ML outputs will be presented as indicators that can help prioritize cases for professional review.

The project meets the requirements of a Software Engineering course by emphasizing:

- Requirements and workflow design
- Role-based access control
- Web architecture
- Database design
- API development
- Testing
- CI/CD
- Deployment
- Scalability
- Measurable evaluation
- Machine-learning integration
- Data privacy
- Human-in-the-loop decision making

The project will follow an iterative development process, beginning with the core recovery-management workflow and progressively adding analytics, alerts, machine learning, and professional dashboards.

The final system will demonstrate how a relatively lightweight machine-learning component can be integrated into a complete software product to transform longitudinal recovery data into useful, explainable, and actionable information for professional review.